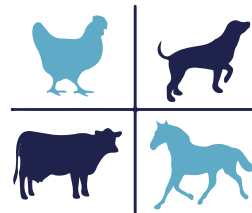


Feed Supplements





MakZyme-C_{AMF}

(Advanced Microchipping Technology)



MakZyme-C_{AMF}

(Advanced Microchipping Technology)



Makzyme-C is a perfect blend of substrate specific Enzymes to enhance the availability of nutrients. Makzyme-C reduces the gut viscosity, improves nutrient digestibility & releases hidden nutrients in feed ingredients.

COMPOSITION

MAKZYME - C AMF Content	
ENZYME	Minimum Activity Unit / kg
Cellulase	63,00,000
Xylanase	1,00,00,000
Beta - glucanase	7,50,000
Phytase	6,00,000
Alpha-amylase	8,00,000
Pectinase	80,000
Protease	30,00,000
Lipase	7,000

ADVANTAGES

- Advanced microchipping form, this technology enables the large surface area of the enzyme while acting in the substrate.
- The stability of Makzyme-C AMF enzyme additives is much better Because enzymes with inherent high stability are selected. New carrier materials are employed & novel manufacturing techniques used.
- Makzyme-C AMF contains high level of carbohydrases complex especially enzyme XYLANASE at high unit's level @ 500 gm/ton, no need to add xylanase separately.
- It's high carbohydrases concentration is used to formulate low cost feed with 25-50 kcal energy replacement & 0.5% of crude protein - @ 500g/ton dosage based on substrate
- High level of thermo stability (even after pelletization 90-95% recovery)



BENEFITS

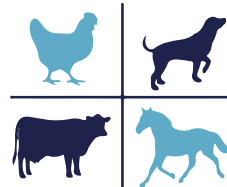
- Allows incorporating alternative feed ingredients & reducing feed cost. Degrading NSPs & improves FCR & production by reducing gut viscosity. This in turn contributes to greater nutrient absorption, better intestinal villus function & health
- **Reduces wet litter problems by neutralizing anti nutritional factors like arabinoxylans, beta-glucans, gossypol, phytate, saponin, tannins & trypsin inhibitors.**
- **Improves nutrient digestion & help to reduce litter problems & reduces ammonia excretion & thereby reduces disease incidences.**
- **Makzyme-C AMF reduces the water holding capacity of the gut contents thus increasing the dry matter content.**
- Improves Hen day & hen Housed Egg production by improving absorption.
- Improves Eggshell Quality due to Phytase Activity.
- Reduces feed cost per egg due to production increment.
- Reduces Dirty Eggs by reducing Gut Viscosity and Wet litter issue in high Fibre Diet.
- Sustains & Improve Production performance during Raw material Quality Fluctuations.

MODE OF ACTION

- **XYLANASE:** It breaks down NSPs, specifically arabinoxylans, which in turn, reduces digesta viscosity, releases trapped nutrients & improves feed passage rate. The released arabino-oligosaccharides may also have a prebiotic effect in the gut.
- **PROTEASE:** Improves the digestibility of protein & reduces anti-nutritional factors, e.g. residual trypsin inhibitors in soybean meal.
- **PHYTASE:** Is capable of breaking down phytate & release phytic acid efficiently, also release phosphorus & other nutrients from phytate.
- **LIPASE:** Breakdowns the lipids from oils in poultry feed, which is more deficit in chick stage.
- **Cellulase:** breakdowns cellulose in fibrous raw material.
- **Beta-mannanase**-breakdowns mannans especially in soybean meal & eliminate the adverse effects.
- A perfect blend of amylase, xylanase, & protease enzymes in Makzyme-C AMF working together synergistically each other and acts on poorly digestible portions of feed ingredients & increases the availability of nutrient for better growth & egg production.

MakZyme-C^{AMF}

(Advanced Microchipping Technology)



Effect on Makzyme-C on Broiler Performance and Intestinal viscosity :

Groups	Body Weight (Gms) @ 34 Days	FCR	Intestinal Viscosity (Cp)
Control	2.094	1.68	1.87
Only Cocktail Enzyme-500gms/Ton	2.103	1.66	1.45
Makzyme - C AMF 350gms/Ton	2.115	1.62	1.32

COMMERCIAL LAYER PERFORMANCE AT 1 MILLION BIRDS META-ANALYSIS

Trail Age- 25 wks 60 wks			
Parameters	Control ~ without Enzyme	Competitor Enzyme at 500gms/ton	Makzyme @ 350gms /ton
Daily Production - Farm Average	88.3%	91.8%	93.8%
Avg. Feed intake in gms/Day	112	111.4	110
Issues	- Dirty Eggs - Loose Droppings - Pasty Vent - Production Fluctuations - Fly Issue - Up to 3% Broken	- Dirty Eggs - Mild Loose Droppings - Production Fluctuations	- Healthy Gut - Gut Scoring mostly 1 - Uniform Production

MATRIX VALUE

- ME: 25-50 Kcal/kg
- Protein: 0.5%
- Replaces: 0.75-1 kg DCP

USAGE

- Makzyme-C AMF is available in **Micro Chip Form**.
- Broilers, Breeders and Layers : 350 gm / ton for regular use.
or As advised by the Veterinarian / Nutritionist.
- For Matrix value 500 gm / ton of feed

PRESENTATION

25kg bag.

SHELF LIFE

24 Months from Date of Manufacture



(An ISO 9001 : 2015 Certified Company)



GMP & HACCP Certified

Marketed By :

ZENMAK NUTRIGENCIES AND HEALTH PVT. LTD.

Regd. Off: No. F2, Sand Stone Apartments,
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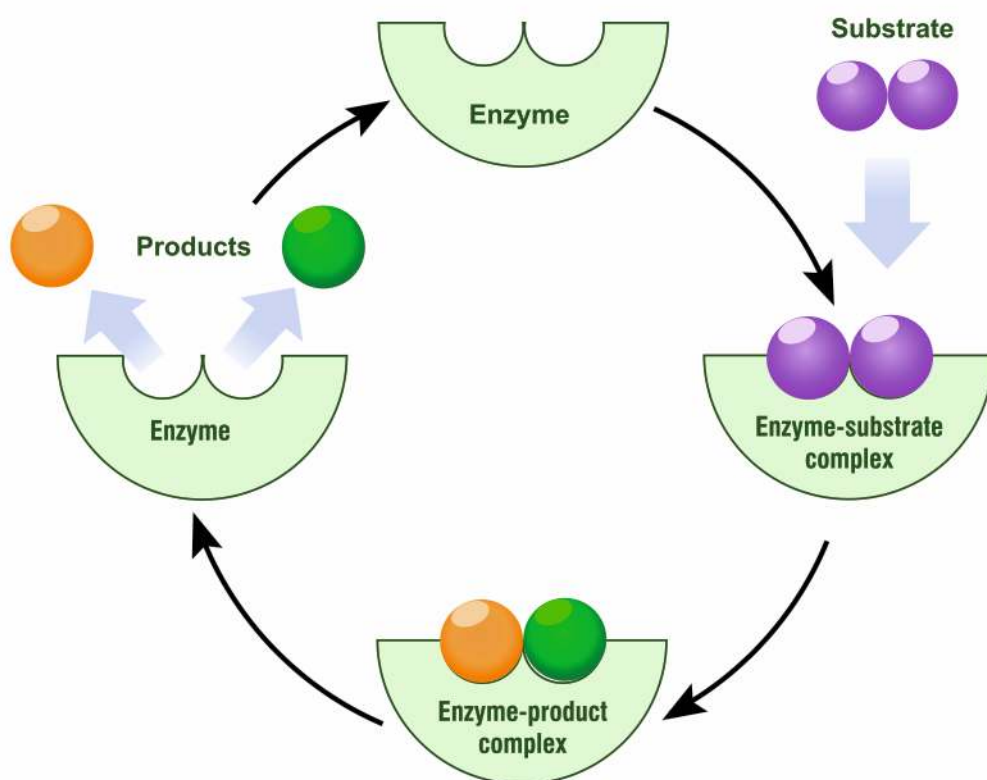
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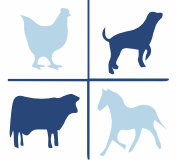




MakZyme- XPL

Premium





Thermostable Extra Strength Xylanase Complex

MakZyme-XPL is a unique enzyme feed additive scientifically developed for consistent & premium performance.

MakZyme XPL Premium is a perfect blend of substrate specific enzymes in powder form to enhance the availability of nutrients especially carbohydrates and protein. MakZyme XPL Premium reduces the gut viscosity, improves nutrient digestibility and releases the hidden nutrients in feed ingredient.

COMPOSITION

- Each Gram Contains
- 1,4 endo xylanase- 1,80,000 BXU/g
- Protease- 40,000 u/g
- Amylase- 2,000 u/g
- Carrier- Starch

1,4 ENDOXYLANASE

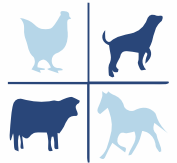
- 1,4 endo-xylanase is used to breakdown xylan, which is a non-starch Polysaccharide found in corn and other grains. Xylanase enzymes hydrolyze glycosidic bonds in xylans. The addition of enzymes to make up to 30% of nutrients more digestible and release of encapsulated nutrients. The additional benefit of xylanase is changing the viscosity due to arabinoxylans and anti-nutritional factors.

Non Starch Polysaccharides Content of Raw Material

Raw Material	Arabino xylans	Beta Glucans	Mannan	Cellulose	Oligosac carides	Others	Total NSP
Corn	4.3	0.3	0.3	2	0.8	0.6	8.3
Wheat	7.1	0.6	0.3	1.8	0.1	0.1	10
Sorghum	3.7	0.1	0.1	1.1	0.2	0.3	5.5
Soybean Meal	0.4	0.7	1.3	9.1	9.6	4.6	25.7
Canola Meal	0.3	0.8	0.5	7.1	3.7	5.6	18
Corn DDGS	11.7	-	0.5	2.7	0.2	10.2	25.3

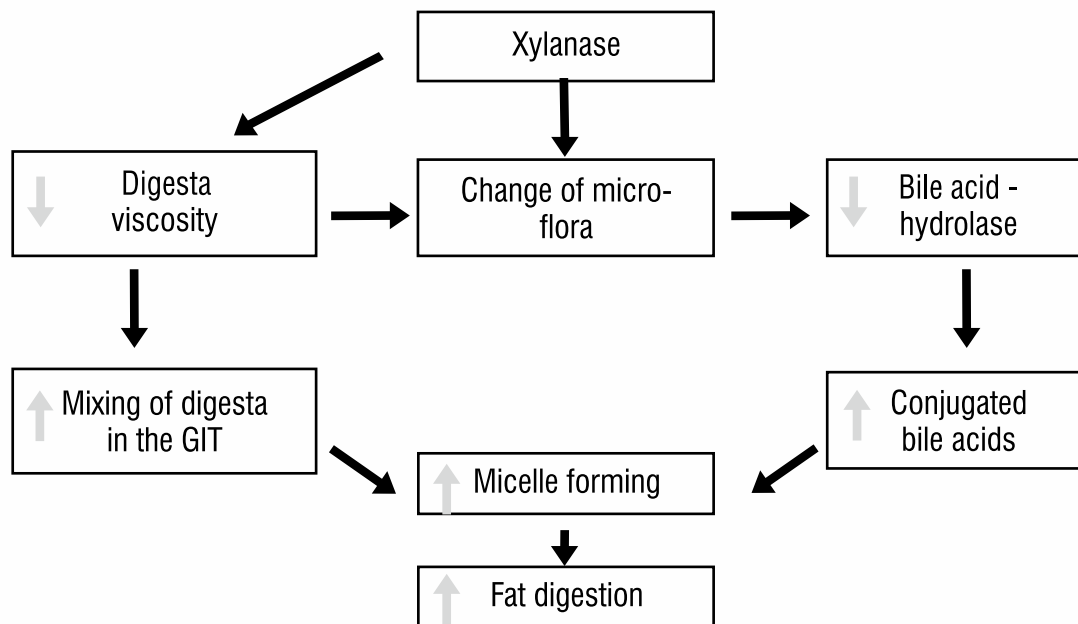
Poultry birds do not produce enzymes for digesting NSP in feed raw materials.

Antinutritional factors (ANF) in NSP cage nutrients & increase the viscosity of digestive tract.



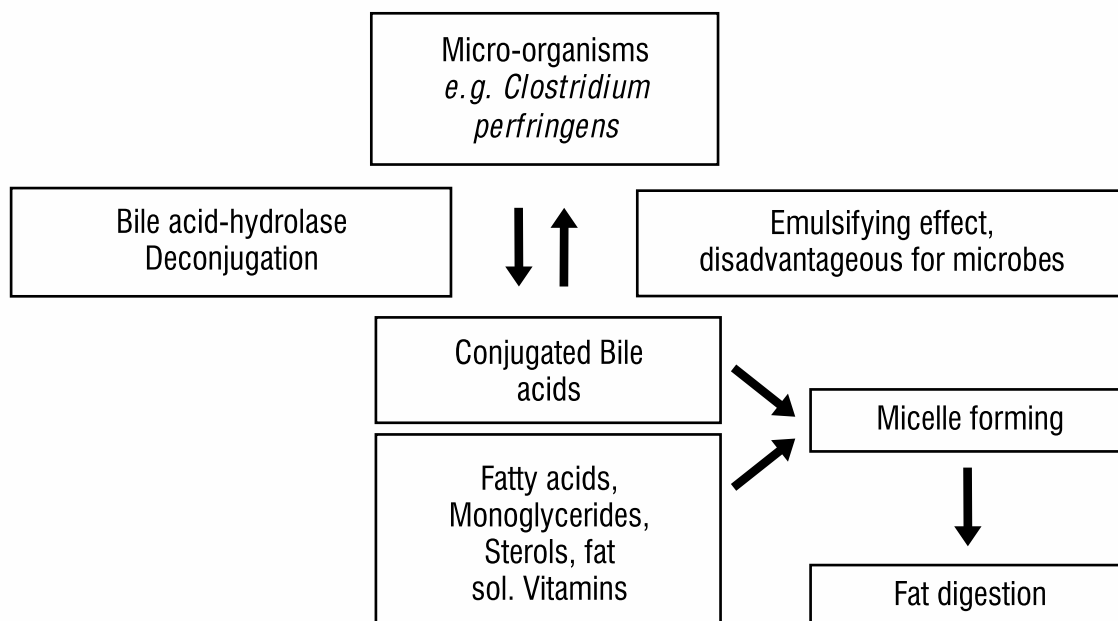
- The interaction between Xylanase supplementation viscosity and fat digestion has been adequately researched. It is known that enzyme supplements, altering substrate availability in the small intestine modify the microflora. Xylanase supplementation reduces the production of bile acid hydrolases by altering the microbe population in the gut. This might result in more conjugated bile acids becoming available and marked the beneficial effect of Xylanase on fat digestibility.

Relation of xylanase and fat digestion (adaptated from HÜBENER *et al.*, 1999)



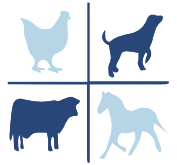
- Long-chain arabinoxylan molecules are breakdown by Xylanase and produce short-chain compounds like Xylo Oligomers which act as a prebiotic. Xylo Oligomer is a nutrient source for beneficial bacteria.

Negative influence of some micro-organisms on fat digestibility. (Cole & Fuller 1984)



MakZyme- XPL

Premium



PROTEASE

Protease is a protein digesting enzyme that breaks down storage proteins binding starch within feed ingredients. This makes the energy from protein bound starch available to the bird to be used for productive purposes. Proteases are also effective in releasing protein anti-nutrients found in ingredients like soybean meal. This function of proteases makes proteins more available. Increases protein digestibility through hydrolysis of storage and structural proteins, and disrupts interactions of proteins with starch and fibre in the diet. Additionally, they target other antinutritional factors in the diet e.g. residual trypsin inhibitors and lectins in soybean meal and some other vegetable proteins, thereby improving nutrient digestibility.

AMYLASE

Acts on starch, increasing its hydrolysis and thereby improving its digestibility by hydrolysis of 1,4 glucosidic linkages. Its actions complement the secretion of endogenous amylase by the bird, freeing up energy to fuel growth. Increasing starch digestibility also reduces the presence of glucose as a potential substrate for non-beneficial bacteria in the latter part of the GIT.

BENEFITS

- Alteration in feed passage rate-MakZyme-XPL Premium reduces the water holding capacity of the gut contents thus increasing the dry matter content stimulating feed intake.
- Reduced viscosity improves nutrient digestion, lower water intake and help to reduce litter problems.
- Addition of MakZyme-XPL Premium also reduces the dry matter outside the body thus has marked impact on excreta volume and composition. Due to protein digestibility lowering of excreta output is observed.
- MakZyme-XPL Premium help to release Nutrients:
 - a) Enzymatic depolymerization renders the NSP free releasing nutrients in the gut and its availability to the animal.
 - b) Starch masked by cell structure is released leading to increase in metabolizable energy.
 - c) Proteins are also released by action of proteases. This release leads to maximum absorption of minerals and results in improvement in nutrient utilization.

FEEDING DIRECTIONS

Broiler/Layers & Breeders: 100 gm/ton of feed
(or) as advised by the Veterinarian

STORAGE

Away from moisture, Hot environment & sunlight

PRESENTATION

25 kg Bag

SHELF - LIFE

24 months from the date of manufacture



(An ISO 9001 : 2015 Certified Company)



GMP & HACCP Certified

Marketed By :

ZENMAK NUTRIGENCIES AND HEALTH PVT. LTD.

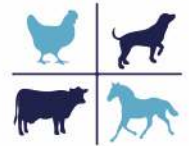
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SAL-O-MAK *Plus*

Potentiated Acidifier for Breeders



SAL-O-MAK *Plus*

Potentiated Acidifier for Breeders



SAL-O-MAK *Plus* is a perfect blend of buffered organic acids mixture, specialized for Breeder flocks, strengthened with Essential oils and Coated Butyric acid. Organic acids have multi usages in poultry feeds as they help in control of microbial growth, reduction of feed buffer capacity, inhibition of pathogenic bacteria and betterment of nutrient digestibility.



COMPOSITION

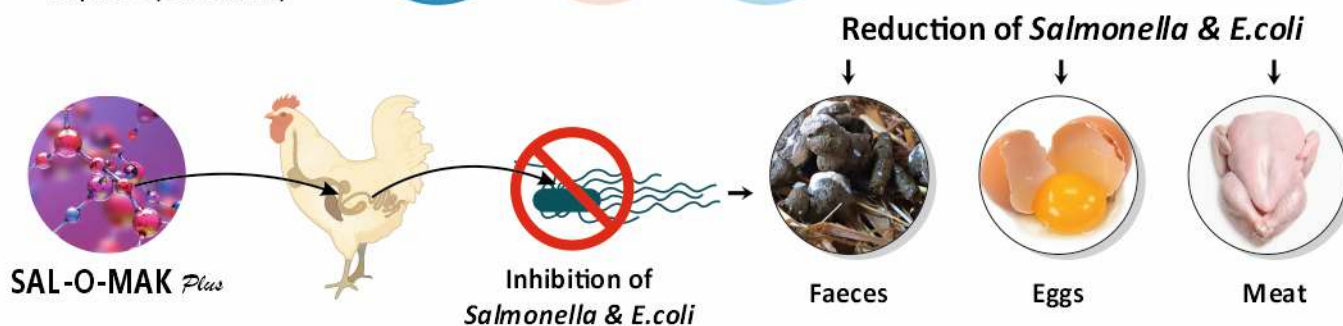
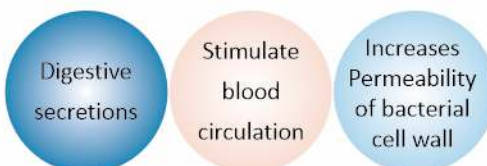
Salts of Citric acid, Propionic acid, Formic acid, Acetic acid, Fumaric acid, Benzoic acid, Lactic acid, Coated Butyric acid and Essential oils.

COATED BUTYRIC ACID

- Gut empowerment : Preferred & high energy source for cells of villi, Increases villi length, Intestinal barrier , Delays feed passage
- Regulates gut microflora
- Inducer of immune cells

ESSENTIAL OILS:

- Plant derived
- Improve palatability



UNIQUE FEATURES

- Less corrosive, less pungent, easy handling in feed mills
- Protection in foregut & Complete release of butyric acid at site of action ie. hindgut
- Concentration of 48-50% of acids
- Broader spectrum of antimicrobial activity

DOSAGE

Breeder : 1 kg - 2 kg/ ton of feed
Broiler / Layer : 500 g/ton of feed
or
Advised by the Veterinarian



PRESENTATION

25 KG bag.

SELFLIFE

24 Months from Date of Manufacture



(An ISO 9001 : 2015 Certified Company)



GMP & HACCP Certified

Manufactured and Marketed By :

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SAL-O-MAK

Broad Spectrum Feed and Gut Acidifer



SAL-O-MAK

Broad Spectrum Feed and Gut Acidifier



Sal-O-Mak is a perfect blend of buffered organic acid mixture, specialized for poultry & aqua, enhanced with essential oils & esterified coated butyric acid. This addition makes the product effective in the finished feed as well as whole intestinal tract. This contributes to the birds stable feed digestion while allowing important digestive enzymes to be activated faster helps in promoting bird's production and growth.

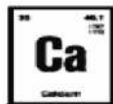
Poultry Feed always contains higher protein values. Protein has buffer effect which absorbs the acids from intestine, which results in higher pH of intestine. With ideal composition of organic acids and advanced blending technology, Sal-O-Mak remarkably reduce the pH of intestine and improves efficacy of enzymes by enhancing release and utilization of nutrients from feed.

COMPOSITION

Salts of Citric acid, Propionic acid, Esterified Coated butyric acid, Formic acid, Acetic acid, Fumaric acid, Benzoic acid & Essential oils

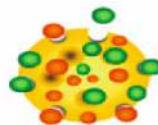
Next generation acidifiers

Combination of 3 (or more) organic acids

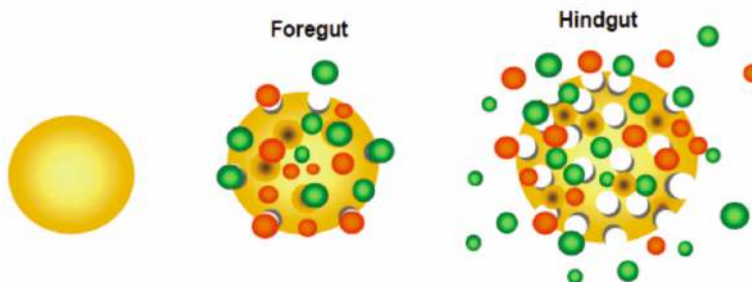


Calcium, Potassium and Sodium salts of Organic Acids have Lower PKa Value

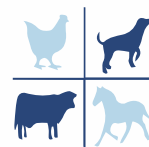
Micro-encapsulation with essential oils



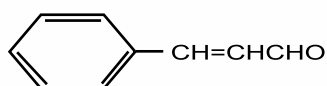
Release of micro-encapsulated organic acids and essential oils



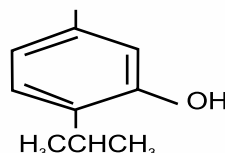
Fat capsule digestion through Lipase



Essential oils are secondary plant metabolites. EOs enhance production of digestive secretions, stimulate blood circulation, exert antioxidant properties, reduce levels of pathogenic bacteria by increase permeability of the cell wall of pathogenic bacteria and enhance immune status.



Antimicrobial
Antiinflammatory
Antispasmodic
Antiurease
Cancer-preventive
Flavor
Hypoglycemic
Cinnamaldehyde



Antimicrobial
Antiinflammatory
Antimelanomic
Antioxidant
Antiseptic
Carminative
Flavor
Thymol

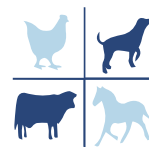
BENEFITS

- **To Control: *E. Coli*, *Salmonella*, *Clostridia* etc.**
Sal-O-Mak's non-dissociated organic acids can penetrate the bacterial cell wall and destroy the normal physiological functions of pH sensitive bacteria.
- **Improve Nutrient Digestibility:**
Digestibility and availability of nutrients (such as Ca and P) due to developing beneficial microflora (*Lactobacillus* spp.)
- **Esterified Coated Butyric Acid in Sal-O-Mak:**
In coated form enables Sal-O-Mak to act effectively in hind gut. Increases the villus width, height and area of the duodenum, jejunum and ileum of broiler chicks.
- Some acids are in liquid form and highly pungent & irritant acids. Sal-o-mak has it, in **sustained release micro pellets** makes it less pungent & easy for handling in feed mills
- Stimulation & activation of digestive enzymes by lowering gut pH & induces the release of digestive enzymes
- Better villi development & aids in absorption of nutrients which leads to appreciable improvement in production parameters like FCR & body weight

ADVANTAGES

- Sal-O-Mak is a Proprietary blend of buffered **organic acid** with **Essential oils** and **Esterified Butyric acid**.
- Micro-Encapsulated Acidifier.
- Does not damage other ingredients in feed.
- Sal-O-Mak is highly diffusive and less corrosive for machineries due to its patented manufacturing technology.
- **Sal-O-Mak's acid concentration is 48-50%**
- Due to its synergistic action of Acidification with Essential Oil, its effective against *Salmonella*, *E.Coli*, *Clostridium* spp., *Pasteurella* etc..





Chemical properties of selected acids and salts

Organic acid	Salts	pk Value	Solubility	Molecular weight (/mol)	Gross energy (MJ/ kg)
Formic acid	Calcium formate, Potassium formate, Sodium formate	3.75	Very good	48.0	5.8
Acetic acid	Sodium acetate	4.76	Very good	60.1	14.8
Propionic acid	Calcium propionate	4.88	Very good	74.1	20.8
Lactic acid	Calcium lactate	3.90	good	90.1	15.1
Fumaric acid	Magnesium fumarate	3.04/4.44	low	116.1	11.5
Citric acid	Calcium citrate, zinc citrate, Sodium citrate	3.14/4.76/6.40	good	210.1	10.3

HOW DO SAL-O-MAK WORKS?

The antibacterial activity of Sal-O-Mak is related to the reduction of pH, as well as their ability to penetrate pathogenic cell wall. In the dissociated form organic acids are lipids soluble in which they are able to enter microbial cell. Once an acid enter into the cell wall releases the proton in more alkaline environment, reduces the intercellular pH. This influences microbial metabolism, leading to an intercellular accumulation of acids actions. The acid anion seems to be very important for the antibacterial effect.

The antibacterial effect of dietary organic acids in chicken is mainly take place in crop & gizzard. Where as Sal-O-Mak blend with esterified butyric acid along with essential oils act also well in small and large intestine of GIT.

APPLICATION

- Feed Preservation as mould inhibitor.
- Gut PH Reduction as gut acidifier.
- Better villi development and aids in absorption of nutrients.
- Pathogenic bacterial control.

DOSAGE

- Broiler / Layers : 500 g - 1kg / ton of feed
- Breeders : 1kg - 2kg / ton of feed

PRESENTATION

25kg bag.

SHELF LIFE

24 Months from Date of Manufacture



(An ISO 9001 : 2015 Certified Company)



GMP & HACCP Certified

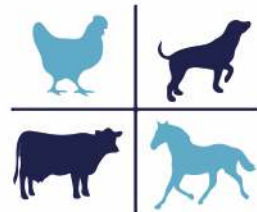
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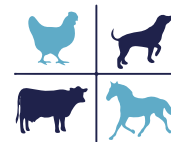




SAL-O-ZEN

A Unique Acidifier





Sal-O-Zen is a perfect blend of buffered organic acid mixture, specialized for poultry. The addition of organic acid mixture brings effectiveness in the finished feed as well as whole intestinal tract. With ideal composition of organic acids and advanced blending technology, Sal-O-Zen remarkably reduces the pH of intestine and improves efficacy of enzymes by enhancing the release and utilization of nutrients from feed.



COMPOSITION

Salts of Citric acid, Propionic acid, Formic acid, Acetic acid, Fumaric acid, Benzoic acid, Lactic acid & Coated Butyric acid.



BENEFITS

- In Sal-O-Zen, the presence of Propionic acid in Ammonium dipropionate form, performs to be the most effective mold inhibitor in feed.
- Sal-O-Zen's non-dissociated organic acids can penetrate the bacteria cell wall and destroy the normal physiological functions of pH sensitive bacteria.
- Better villi development & aids in absorption of nutrients which leads to appreciable improvement in production parameters like FCR & body weight.
- Sal-O-Zen is highly diffusive and less corrosive for machineries due to its patented manufacturing technology.
- Due to its synergistic action of acidification it is effective against *Salmonella*, *E.Coli*, *Clostridium spp.*, *Pasteurella etc.*,
- The selected pKa value of different organic acids, upholds Sal-O-Zen's optimum action in finished feed and intestinal tract of poultry birds.



APPLICATION

- Feed Preservation as mold inhibitor.
- Better villi development and aids in absorption of nutrients.
- Sal-O-Zen favourably manipulate the intestinal microbial population & improve the immune response.
- Performs an activity similar to Anti-microbials in counter acting pathogenic microorganisms.



DOSAGE

- Broiler / Layers / Breeder : 1 kg - 2 kg/ ton of feed
or
Advised by the Veterinarian



PRESENTATION

25 KG bag.

SELF LIFE

24 Months from Date of Manufacture



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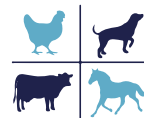
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Lipolyse-C

The Most Potent Hydrophilic Emulsifier

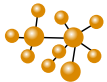


The Most Potent Hydrophilic Emulsifier

OIL INCLUSION IN BROILER IS IMPORTANT, WHY?

Lipids constitute the main energy reserve of birds and it has the highest caloric value among all nutrients. The Carbon atoms of the fatty acids are chemically more than C atoms found in Sugar. Therefore, the oxidation of triglycerides releases more than twice as much energy as Carbohydrates.

The deposition of 1 g of energy from carbohydrate (or) protein by a poultry bird requires higher quantities of these nutrients in comparison to the deposition of 1g of energy from fat.



COMPOSITION

- GPGR (Glyceryl Poly Ethylene Glycol Ricinoleate)
- Lysophospholipid
- Phospholipids
- Soluble Caseinates
- Lipase
- Synthetic Bile Salt
- Natural Carriers



HOW DOES AN EMULSIFIER WORKS?

Emulsifier molecules work by having a hydrophilic end (water-loving) and hydrophobic end (water-hating). The hydrophilic end of the emulsifier molecule is attracted to the water and the hydrophobic end is attracted to the fat/oil. By vigorously mixing the emulsifier with the water and fat/oil, a stable emulsion can be made.

Fat emulsification in the digestive tract occurs with the help of bile salts and later digested by Lipase. But, the most critical step is digestion of Fatty acid containing Carbon atoms >14. As most of the oil included in the broiler diet has fatty acid with >14 Carbon atoms, further addition of an emulsifier in feed becomes mandatory .

EMULSIFIERS ARE REQUIRED IN :



YOUNG BIRDS: Young birds have limited ability to digest fats which is due to limited bile salt secretion and absorption, which increases only with age. So, In broilers the addition of emulsifiers plays major role in production cost .

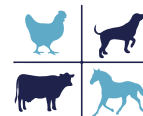
DISEASE CONDITION: Pathogenic organism affects intestinal epithelium. Epithelial cells are ruptured by which absorption of essential nutrients are decreased.



CEREALS HAVING HIGH LEVEL OF NSPs: High level of NSPs increases the viscosity of digesta in chicken intestine and interferes in absorption of nutrients. Increased viscosity also leads to impaired peristaltic progress of chyme through intestine which leads to bacterial overgrowth, causing loss of conjugation of bile salts and reduction of fat utilisation.

All the above factors causes poor growth in broiler chickens with great loss to farmers . So, it becomes mandatory to add emulsifiers in the diet of broiler ration.

Glyceryl Poly Ethylene Glycol Ricinoleate (GPGR)



- ✦ More hydrophilic and dissolves the free FA which are hardly soluble in bile salt, enhances dietary fat availability, improves body weight and relative organ weight.
- ✦ GPGR maintains serum and meat lipid profile.
- ✦ Adding GPGR in feed before pelleting increases humidity and decreases pellet press energy consumption and improves the pellet quality by modulating the moisture content during pelleting process.

LYSOPHOSPHOLIPID or LYSOLECITHIN:

- ✦ Lysophospholipid improves body weight gain, FCR, improves carcass quality and dressing percentage. Choline is mainly meant to reduce fatty liver condition in birds and promotes jejunal function in broilers.
- ✦ It acts as an agonist of PPAR α (Peroxisome Proliferator Activated Receptors Alpha), proteic nuclear receptors acting on transcription process. The PPAR α regulate the expression of specific genes implicated in metabolic reactions, such as β -oxidation of fatty acids.
- ✦ Lysophospholipid are modulators of epigenetic regulation of gene expression, which improves fatty acid digestion in the birds.
- ✦ Acts as a methyl group donor via betaine, helps in synthesis of phospholipids and maintains cell wall integrity.

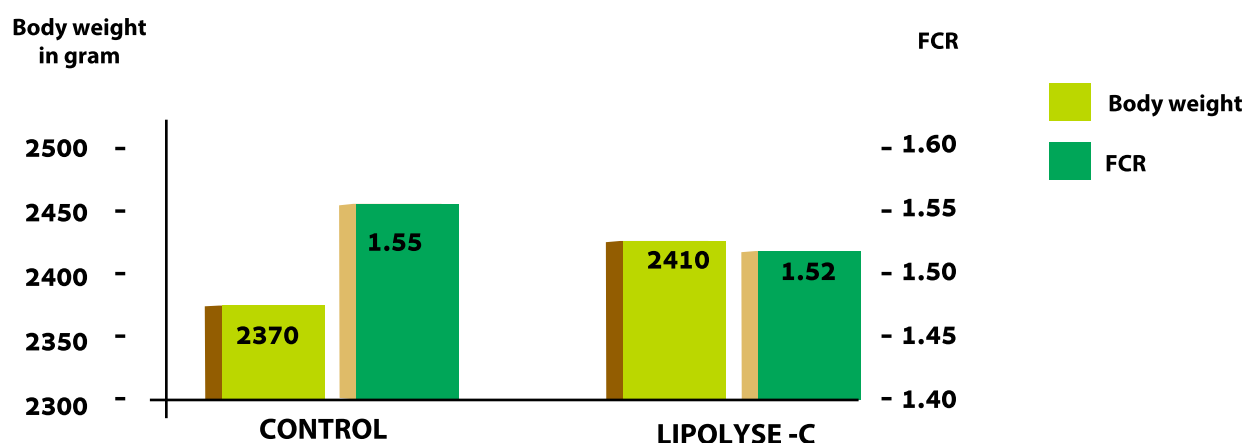
SOLUBLE CASEINATES:

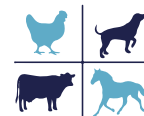
- ✦ A potent emulsifying agent due to its amphiphilic property. It combines with both water and oil phase and stabilizes the emulsion. Increases absorption of proteins & fatty acids, improves body weight gain and increases lipase activity.

SYNTHETIC BILE SALT

- ✦ Added Synthetic Bile Salt in LIPOLYSE-C activates lipase enzyme, increases ME, improves digestion of fat molecules, decreases plasma cholesterol.

Effect of LIPOLYSE -C supplementation on FCR & body weight of broilers at 42 days of age





LIPASE

Lipases are defined as triacylglycerol acyl hydrolases that are involved in the hydrolysis of fats and oils to yield glycerol and free fatty acids.. Young birds with immature digestive capability for complete absorption of lipids could particularly benefit from dietary lipases (Pleiss et al., 1998; Fickers et al., 2011).

HYDROPHILIC LIPOPHILIC BALANCE:

The extent to which the part of an emulsifier are hydrophilic & lipophilic in nature determines HLB value. HLB is the key indicator for selecting an emulsifier, ranging from 0-20 . As, in the gut it is Oil-Water emulsion high HLB is required.



KEY BENEFITS:

- Increases the digestibility of fat & oil in the feed.
- Increases the digestion & absorption of proteins in the diet.
- Facilitate low cost feed formulation.
- Minimizes abdominal fat .
- Enhances the secretion and activities of lipase.
- Maximizes enzymatic digestion of feed.
- Suitable for pelleting.



DOSAGE

Broiler/Layer : 500 g/ton of feed.

or

Advised by the Veterinarian



PRESENTATION

25 KG Bag



SHELF LIFE

24 Months from Date of Manufacture.

STORAGE

Keep it in a cool place.



(An ISO 9001 : 2015 Certified Company)



GMP & HACCP Certified

Manufactured and Marketed By :

ZENMAK NUTRIGENCIES AND HEALTH PVT. LTD.

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Contact : 93459 13689, 93449 31464

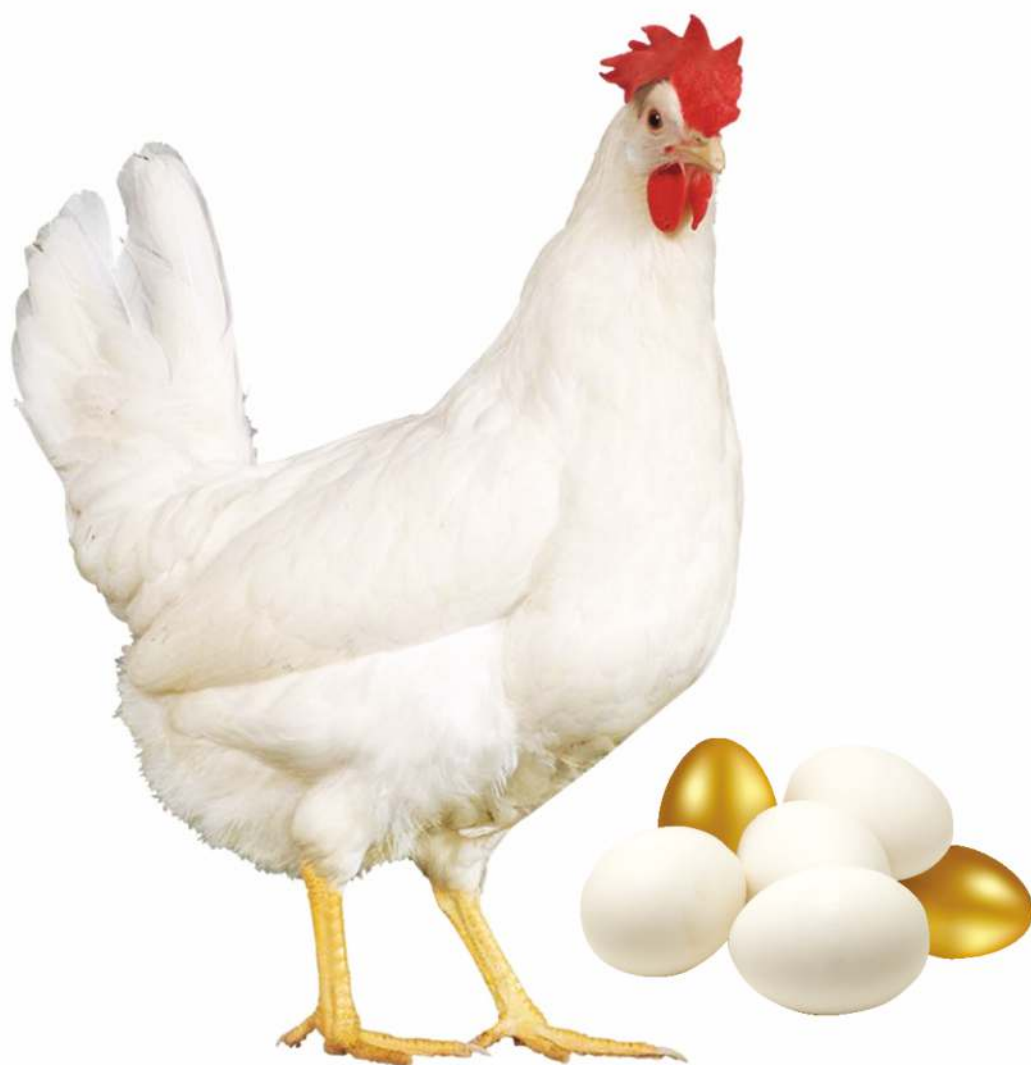
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Lipolyse-L

Effective Fat Mobiliser



Lipolyse-L

Effective Fat Mobiliser



Fat mobiliser are used in poultry nutrition especially in layers for improving fat metabolism, improves laying capability and enhances digestibility. The selection of natural & synthetic sources is based on their activity in fat mobilization process.

DESCRIPTION

In layers after the age of 40 weeks, the abdominal fat deposition increases. The increased abdominal fat leads to the risk of prolapse in layer birds, due to which egg production decreases. Hence balanced supplementation of nutrition along with Choline chloride, Vit.B12, Vit. C and other synthetic component which is responsible for hydrolysis of lipids & produces fatty acid which are water soluble help the hens metabolize fat more efficiently (Kathy S. Mormino;2014).

Choline is a beta-hydroxy ethyl trimethyl ammonium hydroxide. Choline is required for the formation of a phospholipid lecithin, a component of egg yolk.

Choline plays an essential role in fat metabolism in the liver. It prevents abnormal accumulation of fat (fatty livers) by promoting its transport as lecithin or by increasing the utilization of fatty acids in the liver itself (Xu et al., 2010). Choline is thus referred to as a “lipotropic” factor due to its function of acting on fat metabolism by hastening removal or decreasing deposition of fat in liver, as a constituent of phospholipids, i.e. it is essential in the building and maintenance of cell structure, as well as ensuring normal maturation of the cartilage matrix of bone.

- **High Bioavailability:**

Bioavailability is the most important parameter, which should be taken into account while selecting an ideal choline chloride.

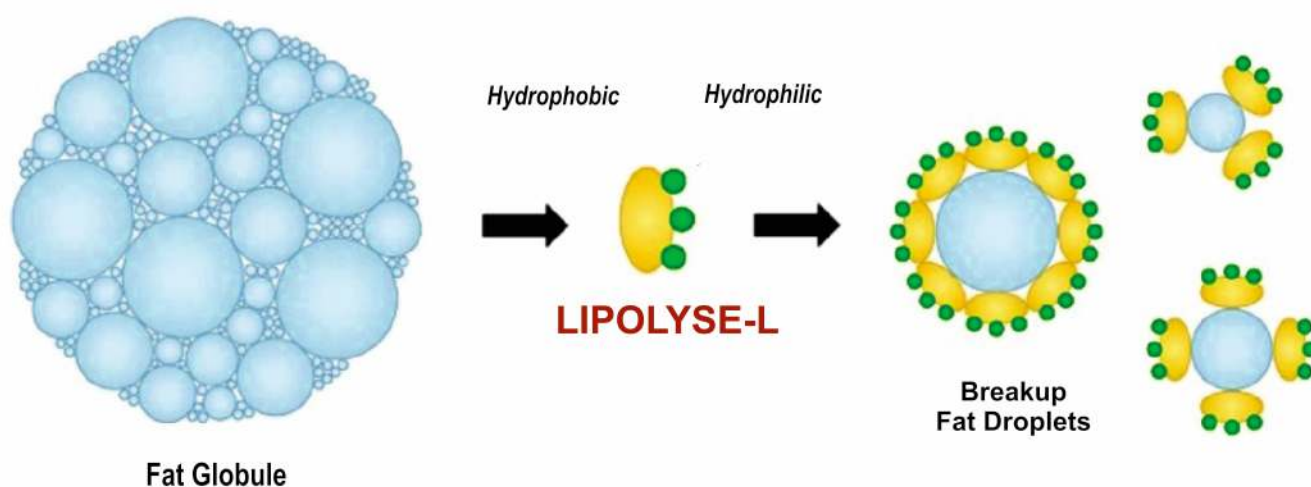
Phosphatidylcholine, one of the active components of natural choline, acts as an agonist (promoter) of the PPAR α (Peroxisome Proliferator Activated Receptors Alpha), proteic nuclear receptors acting on transcription processes. The PPAR α regulate the expression of specific genes implicated in metabolic reactions, such as beta oxidation of fatty acids.

It has been demonstrated that PPAR α are present both in the intestine and in the liver, indicating that natural choline (phosphatidylcholine) begins to act in the enterocytes and, later, in the liver, optimizing the absorption of nutrients in the diets and improving feed conversion and other productive parameters.



Choline is building block of lecithin (phosphatidylcholine). Lecithin is one of the basic constituents of animal cell membranes. Choline is released from lecithin by hydrolysis in the intestinal lumen. Both pancreatic secretions and intestinal mucosal cells contain enzymes capable of hydrolyzing lecithin in the diet. The net result is that most ingested lecithin is absorbed as lysophosphatidylcholine.

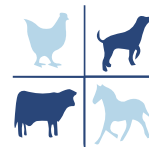
In Lipolyse-L we used natural precursor lysophosphatidylcholine, for choline chloride, which having high bio-availability, work as fat mobiliser and prevents abnormal fat deposition in birds.



- **SOLUBLE CASEINATES:** Increases absorption of protein, increased weight gain, pancreatic lipase secretion, improved fat digestibility with no effects on egg quality.
- **LYSOPHOSPHATIDYLCHOLINE:** precursor of choline chloride, improved body weight, FCR, improve carcass quality.
- **SYNTHETIC SALTS OF ENZYMATIC ACTION :** Lipolyse-L having synthetic salts of enzymatic action. Synthetic salts of enzymatic action helps to digestion of fat molecules, activate lipase enzyme, increases metabolizable energy.
- **GLYCEROL POLYETHYLENE GLYCOL RICINOLEATE:** Enhance dietary fat availability, improved performance, body weight gain & relative organ weight.
- **HERBAL Vit.C:** Helps to improve shell quality, exhibits antioxidative effect, inhibits oxidative protein denaturation.
- **Vit B12:** Initiates nucleic acid & methyl synthesis, helps to lipid metabolism, improve egg weight & hatchability, maintain nervous system integrity.
- **LIVER Rejuvenators :** Improved liver function, increased production of bile for digestion of fat, deposits fat into tissues for further energy utilization.

Lipolyse-L

Effective Fat Mobiliser



COMPOSITION

Soluble caseinates, Lysophosphatidylcholine, Synthetic salts of enzymatic action,
Glycerol polyethylene Glycol ricinoleate,
Herbal vit.C, Vit. B12, Liver tonic & natural carriers.

BENEFITS

- Lipolyse-L contains precursor lysophosphatidylcholine so it easily replaces choline chloride
- Increase the metabolizability of fat in body
- Facilitate low cost formulation
- Minimizes abdominal fat
- Maximizes enzymatic digestion

INCLUSION LEVEL

Layer / Breeder : 1 Kg/ton of feed

Replace 1 Kg of Choline Chloride with 750g of Lipolyse-L

PRESENTATION

25kg bag.



(An ISO 9001 : 2015 Certified Company)



GMP & HACCP Certified

Manufactured and Marketed By :

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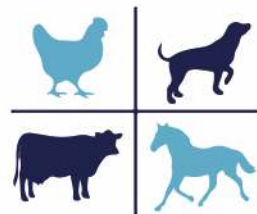
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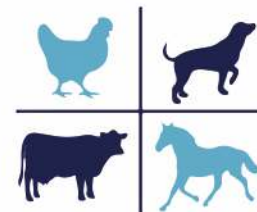
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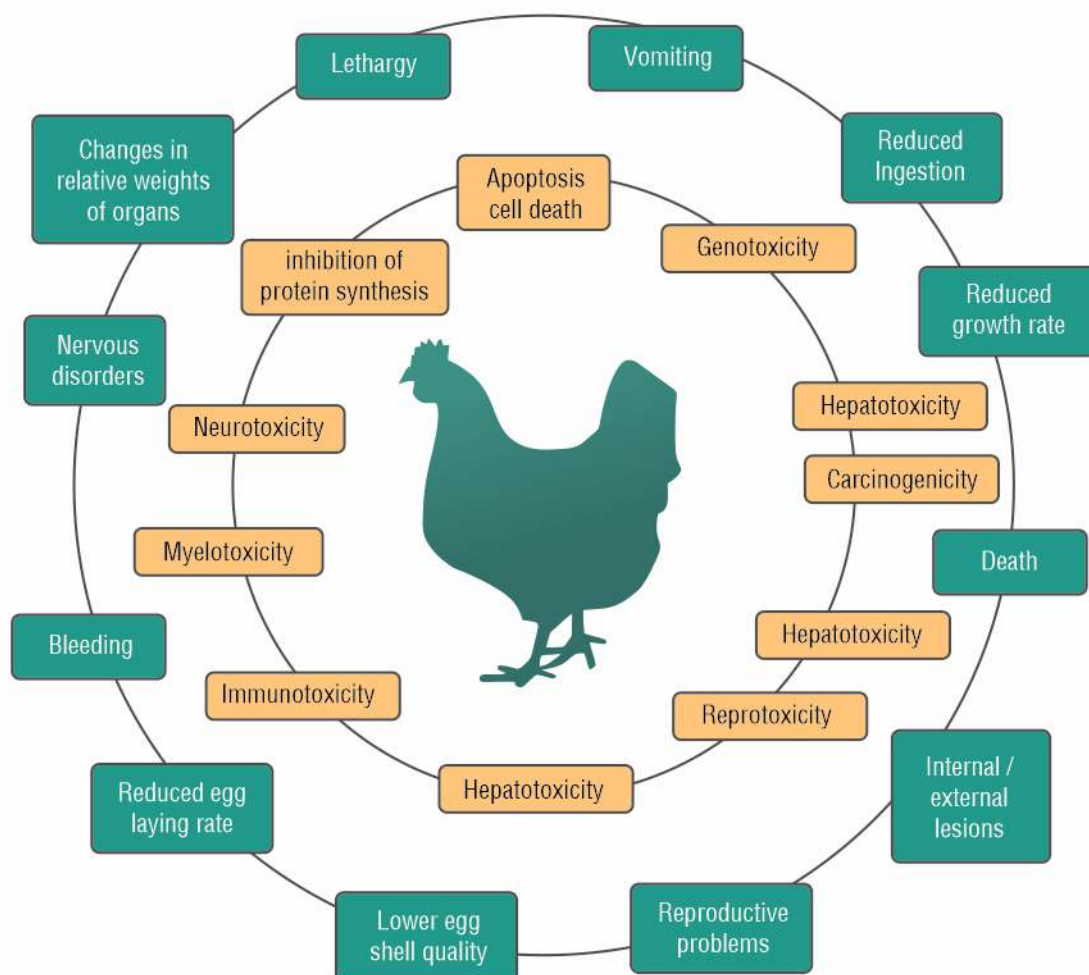
DiTox-3 *Premium*





Is an activated, most effective mycotoxin adsorbent, hydrated sodium calcium aluminosilicate (HSCAS) is specially formulated to adsorb & eliminate all major mycotoxins affecting poultry health & productivity.

EFFECT OF MYCOTOXINS ON POULTRY



PURPOSE : To bind and eliminate mycotoxins & their secondary metabolites (chemicals), reduce pH of feed & gut

COMPOSITION

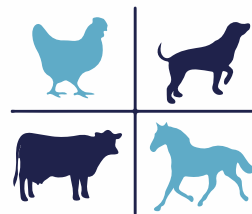
Natural dipolar phyllosilicates, Activated carbon, Mannan oligosaccharides (MOS), Liver & Kidney rejuvenators, Beta glucans & organic acids, Phyllanthus niruri

ACTIVATED CHARCOAL

Used in DiTox-3 premium having properties of high adsorption, low contents of ash, iron, acid soluble, water soluble, and free from heavy metals making it suitable for high rate of mycotoxin adsorption. Activated Charcoal was found to reduce ochratoxin and is efficacious in preventing T2 toxicosis.

NATURAL DIPOLAR PHYLLOSILICATES

Natural dipolar phyllosilicates, are effective toxin binders used in poultry feed to mitigate the harmful effects of mycotoxins like aflatoxins and zearalenone. Their layered structure and dipolar nature enable strong adsorption of polar toxins, preventing their absorption in the gut.



MANNAN OLIGOSACCHARIDES (MOS)

Mannan oligosaccharides (MOS) are natural toxin binders derived from yeast cell walls, known for their dual role in poultry feed. They effectively bind mycotoxins, particularly zearalenone and ochratoxins, reducing their absorption in the gut. MOS also support gut health by inhibiting the binding of harmful pathogens, such as E. coli and Salmonella, to intestinal walls.

ACIDIFIER

Included in feed to reduce the pH of feed, gut and microbial cytoplasm thereby inhibiting the growth of pathogenic intestinal microflora

BETA GLUCANS

Immunomodulator, as substitutes for antibiotics. It is derived from yeast may be ideal for substitutes because of their positive effects on the birds immune system without adversely affecting poultry performance. Improved survival rate & feed efficiency.

PHYLLANTHUS NIRURI

Phyllanthus niruri, is a potent hepatoprotective herb widely used to protect the liver against toxins. Its bioactive compounds, such as phyllanthin and hypophyllanthin, exhibit antioxidant and anti-inflammatory properties, neutralizing free radicals and reducing oxidative stress. The herb enhances liver enzyme function, supports detoxification, and protects hepatocytes from damage caused by mycotoxins. Additionally, it promotes bile secretion, aiding in efficient digestion and toxin elimination.

LIVER AND KIDNEY REJUVENATORS

Helps to keep healthy liver & kidney during detoxifying the toxins

Guideline limits for mycotoxins in Poultry feeds to reduce negative effects on health and performance:

Relative toxicity of different mycotoxins on different livestock species

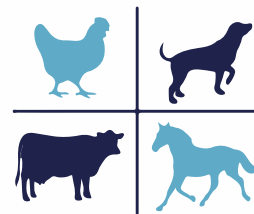
(★ mild toxicity ★★ moderate toxicity ★★★ high toxicity)

	Maximum tolerance limit*
Aflatoxin	20 ppb
Ochratoxin A	40 ppb
Citrinin toxin	100 ppb
Zearalenone	400 ppb
DON (Vomitoxin)	5 ppm
T2 toxin	200 ppb

Toxin	Poultry	Swine	Ruminant
Aflatoxins	★★★	★★	★
Ochratoxins	★★★	★	★
T2 toxin	★★	★★★	★★★
Deoxynivalenol	★	★★	★★
Zearalenone	★	★★★	★★
Fumonisin	★	★★★	★

- DiTox-3 premium have a high affinity for aflatoxin B1 forming a complex which was stable at temperatures of 25 and 37OC. in a pH range of 2-10.
- DiTox-3 premium is used to eliminate Chemical Toxins.

DiTox-3^{Premium}



BENEFITS

1. Binds and eliminate wide range of mycotoxins, chemical toxins and bacterial toxins, ensuring safer and higher-quality feed for poultry.
2. Phyllosilicates adsorb toxins effectively due to their high surface area and layered structure, allowing them to trap and neutralize harmful substances within their interlayer spaces.
3. Organic acids in Ditox-3 premium can neutralize bacterial toxins by lowering pH in the gut, disrupting bacterial cell walls, and binding toxins directly.
4. Improves Liver and Kidney functions.

MAIN EFFECT OF DITOX-3 premium

- Effective degradation, adsorption of mold & its metabolites.
- Optimize the intestinal metabolism, improve digestion.
- Stimulate the body's immune response, enhance immunity, reduce the incidence of a disease.
- Improve bird's production & reproductive performance
- Improving economic efficiency
- Completely replaces liver tonic usage in feed and water.

DOSAGE

Regular : 1 kg/tonne of feed

High moisture and toxin suspected feed : 2 kg/tonne of feed
or As advised by the Veterinarian.

PRESENTATION

25 KG bag

SHELF LIFE

24 months from the date of manufacture.



(An ISO 9001 : 2015 Certified Company)



GMP & HACCP Certified

Manufactured and Marketed By :

ZENMAK NUTRIGENCIES AND HEALTH PVT. LTD.

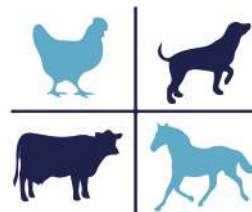
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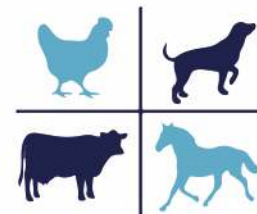




DiTox-3 *Plus*

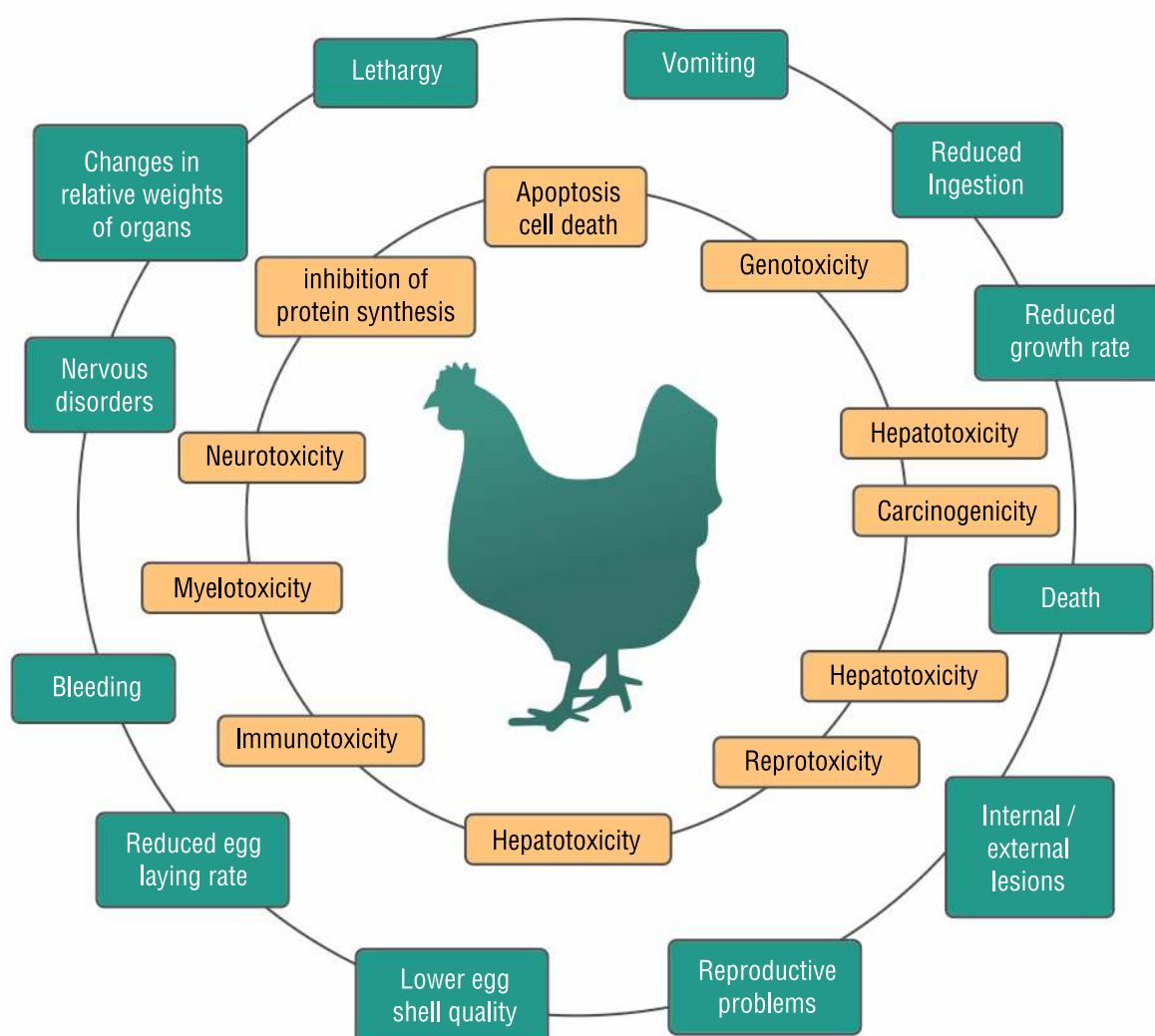
BINDS & ELIMINATE TOXIN





Is an activated, most effective mycotoxin adsorbent, hydrated sodium calcium aluminosilicate (HSCAS) is specially formulated to adsorb & eliminate all major mycotoxins affecting poultry health & productivity.

EFFECT OF MYCOTOXINS ON POULTRY



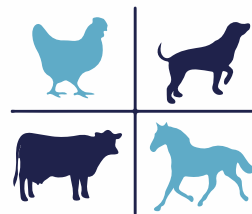
PURPOSE : To bind and eliminate mycotoxins & their secondary metabolites (chemicals).

COMPOSITION

Natural dipolar phyllosilicates, Activated carbon, Mannan oligosaccharides (MOS), Neem powder, Liver & Kidney rejuvenators.

DiTox-3^{Plus}

BINDS & ELIMINATE TOXIN



ACTIVATED CHARCOAL

Used in DiTox-3 plus having properties of high adsorption, low contents of ash, iron, acid soluble, water soluble, and free from heavy metals making it suitable for high rate of mycotoxin adsorption. Activated Charcoal was found to reduce ochratoxin and is efficacious in preventing T2 toxicosis.

NATURAL DIPOLAR PHYLLOSILICATES

Natural dipolar phyllosilicates, are effective toxin binders used in poultry feed to mitigate the harmful effects of mycotoxins like aflatoxins and zearalenone. Their layered structure and dipolar nature enable strong adsorption of polar toxins, preventing their absorption in the gut.

MANNAN OLIGOSACCHARIDES (MOS)

Mannan oligosaccharides (MOS) are natural toxin binders derived from yeast cell walls, known for their dual role in poultry feed. They effectively bind mycotoxins, particularly zearalenone and ochratoxins, reducing their absorption in the gut. MOS also support gut health by inhibiting the binding of harmful pathogens, such as E. coli and Salmonella, to intestinal walls.

NEEM POWDER

Antibacterial, antiviral, antifungal, antiprotozoal, hepatoprotective, immunomodulator.

LIVER AND KIDNEY REJUVENATORS

Helps to keep healthy liver & kidney during detoxifying the toxins

Guideline limits for mycotoxins in Poultry feeds to reduce negative effects on health and performance:

	Maximum tolerance limit*
Aflatoxin	20 ppb
Ochratoxin A	40 ppb
Citrinin toxin	100 ppb
Zearalenone	400 ppb
DON (Vomitoxin)	5 ppm
T2 toxin	200 ppb

Relative toxicity of different mycotoxins on different livestock species

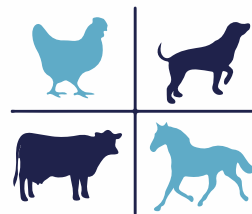
(★ mild toxicity ★★ moderate toxicity ★★★ high toxicity)

Toxin	Poultry	Swine	Ruminant
Aflatoxins	★★★	★★	★
Ochratoxins	★★★	★	★
T2 toxin	★★	★★★	★★★
Deoxynivalenol	★	★★	★★
Zearalenone	★	★★★	★★
Fumonisin	★	★★★	★

- DiTox-3 plus have a high affinity for aflatoxin B1 forming a complex which was stable at temperatures of 25 and 37°C. in a pH range of 2-10.
- DiTox-3 plus is used to eliminate Chemical Toxins.

DiTox-3^{Plus}

BINDS & ELIMINATE TOXIN



BENEFITS

1. Binds and neutralizes toxins, ensuring safer and higher-quality feed for poultry.
2. Mannan oligosaccharides (MOS) attracts and bind pathogens and their toxins in the gastrointestinal tract.
3. Phyllosilicates adsorb toxins effectively due to their high surface area and layered structure, allowing them to trap and neutralize harmful substances within their interlayer spaces.
4. Improves Liver and Kidney functions.

MAIN EFFECT OF DITOX-3 plus

- Effective degradation, adsorption of mold & its metabolites.
- Optimize the intestinal metabolism, improve digestion.
- Stimulate the body's immune response, enhance immunity, reduce the incidence of a disease.
- Improve bird's production & reproductive performance
- Improving economic efficiency

DOSAGE

Regular : 1 kg/tonne of feed

High moisture and toxin suspected feed : 2 kg/tonne of feed

or As advised by the Veterinarian.

PRESENTATION

25 KG bag

SHELF LIFE

24 months from the date of manufacture.



(An ISO 9001 : 2015 Certified Company)



GMP & HACCP Certified

Manufactured and Marketed By :

ZENMAK NUTRIGENCIES AND HEALTH PVT. LTD.

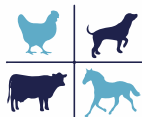
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Organic



Mak Min-OrgTM

More than just a Organic Trace Mineral





The supply of trace mineral in poultry feed is essential for bird health, immune status, allowance for expression of production potential, and improvement in egg shell quality.

Organic trace minerals are used in poultry diets because of their higher bioavailability.

Proteinate Complex

To get the most nutritional and economic value from a feed formulation, Nutritionists and Farmers need to pay special attention to what they are using, especially when it comes to “organic” trace mineral sources.

Choosing the Best Organic Trace mineral for Better ROI:

Peptides, having a greater number of donor atoms which enable their potential to form numerous chelate rings when bonding to a metal ion, would be expected to form more stable chelates than simple amino acids, such as glycine.

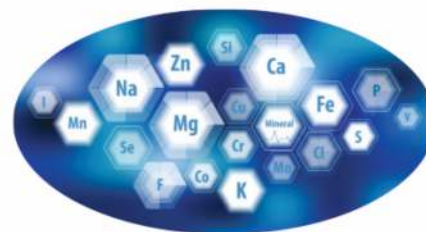
Simply increasing the number of amino acids in a ligand may not increase the stability of the metal complex nor the relative proportion of bound mineral. Ultimately, not only does the amino acid type influence stability but also the amino acid's position in a peptide.

ROLE OF TRACE MINERALS

Trace Minerals	Role of Trace Mineral	Deficiency Symptoms
Zinc	• Essential component of 200 Enzyme systems. • Major role in Immune system and certain reproductive hormones. • Essential for proper sexual maturity and reproductive capacity.	• Result in poor semen quality. • Retardation in growth, leg abnormality with long bones becoming shorter, thickened & crooked. • Hatched chicks are weak with skeletal deformities. • Delayed feathering.
Manganese	• Activator of enzyme system in the metabolism of carbohydrates, fats, proteins and nucleic acid. • Manganese involves in the formation of Chondroitin sulphate & Mucopoly saccharide- bone cartilage formation • Eggshell formation and the function of the immune system.	• Retarded growth • Characteristic crippling leg deformity called perosis. • Reduction in eggshell strength. • Reduction in hatchability.
Copper	• It is a necessary component of number of enzymes which function in increased structural strength, elasticity of connective tissues and blood vessels. • Helps in the maturation of erythrocytes.	• Anaemia, Enlarged heart and fragile bones, leg weakness.
Iron	• Constituent of haemoglobin and myoglobin for oxygen transport and cellular use. • Component of many enzymes such as Cytochrome C, peroxidase, catalase etc.	• Hypochromic microcytic anaemia • Depigmentation of feathers • Stunted growth • Heart hypertrophy
Selenium	• Functions as Antioxidant • Protection of unsaturated tissue lipids against peroxidation. • Plays an important role in fertility with Vitamin E	• Stunted growth, poor feather development and degeneration of pancreas • Exudative diathesis.
Iodine	• It is required for synthesis of thyroid hormone, thyroxin, which regulates the rate of metabolism. • Regulates rate of cellular oxidation. • Regulates neuromuscular functions.	• Increase in the size of thyroid gland called goitre. • Retention of yolk in the hatched chicks. • Lower hatchability.

DIFFERENT FROM OTHER OTM

MakMin-ORG's Production process ensures optimum protein hydrolysis for mineral chelation relies on carefully selecting the hydrolysis conditions, ensuring that the final hydrolysed peptide mix can guarantee constancy and optimal mineral binding stability under conditions of changing pH inside the Gut of Bird.



UNIQUENESS OF MAKMIN ORG

- No Phytase and Xylanase activity reduction like other minerals due to optimum Makmin-ORG stability.
- Due to its optimal proteinate technology, it ensures nil mineral binding with toxin binders and better vitamin stability.
- The key defining characteristic of Makmin-ORG is the ability to maximize its pH-dependent stability and optimize delivery to the intestine.

BENEFITS

In Breeders:

- Anti-oxidant status
- Fertility & Hatchability
- Egg Shell Quality
- Less culls percentage
- Better Chick quality and lameness issue

In Broilers:

- Better Day Gain
- Meat Quality- Low Drip lose
- Less incident of Wooden breast syndrome
- Better FCR and Immune status

In Layers:

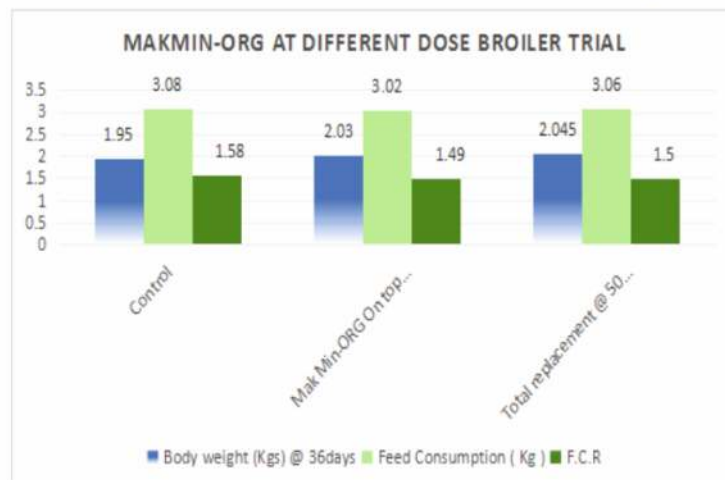
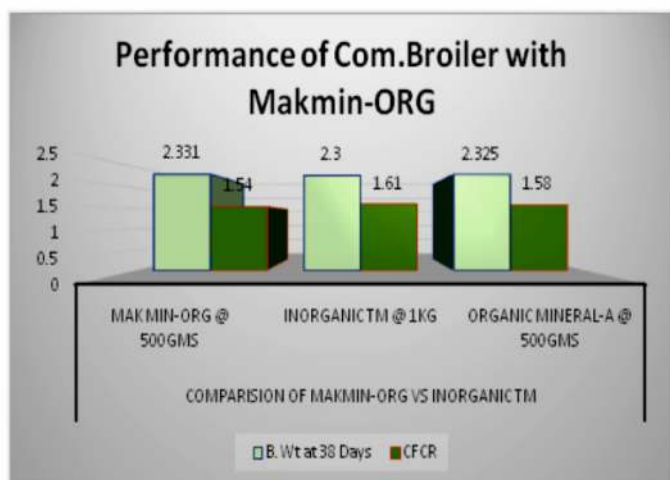
- Shelf life of eggs
- Egg Freshness
- Eggshell quality improvement



ADDED ADVANTAGE

1. Better Gut Health
2. Pathogenic Load Reduction
3. Healthy Feet and Shiny Feather
4. Less Culls Percentage in each batch while lifting

TRIAL DATA



COMPOSITION

NAME OF THE MINERAL	EACH KG CONTAINS (in grams)
Zinc	60.0
Iron	30.0
Copper	10.0
Selenium	0.60
Manganese	60.0
Chromium	1.0
Iodine	4.0

DOSAGE

In Broilers : 500 - 650gms per ton of feed

In Breeders : 1 - 1.2 Kg per ton of feed

* As advised by Veterinarian or Nutritionist



(An ISO 9001:2015 Certified Company)



GMP & HACCP Certified

Marketed By:

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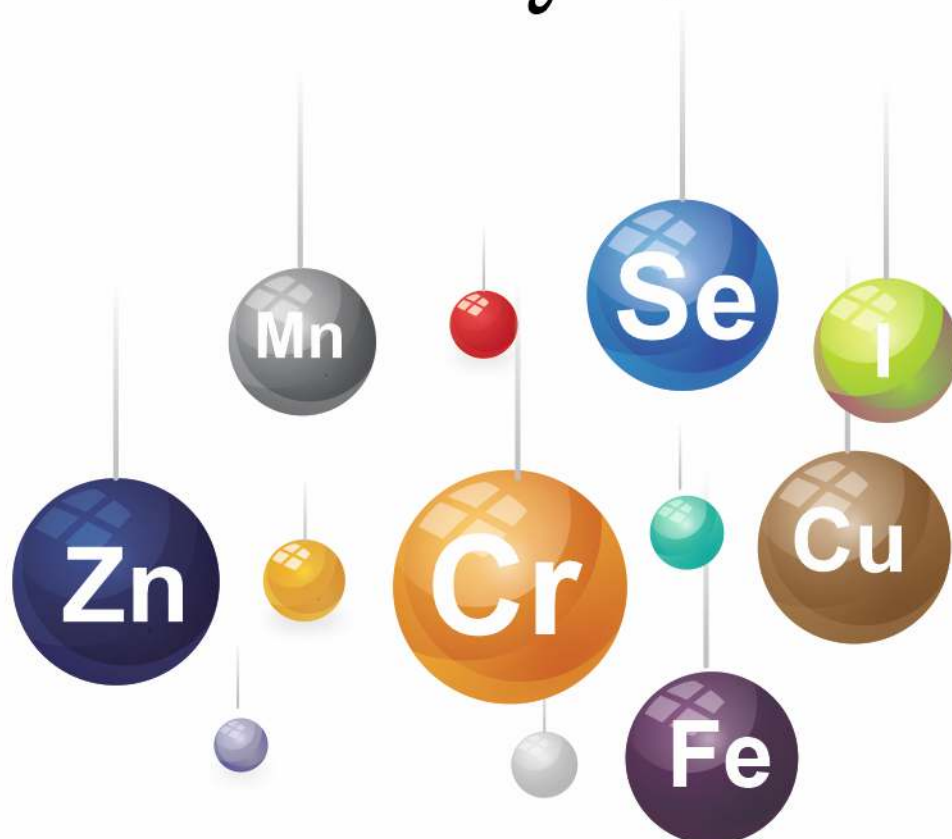
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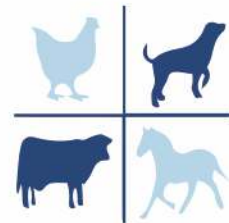
Mak Min-OrgTM

Layer Pack



Mak Min-OrgTM

Layer Pack



Mak Min-Org is a comprehensive mixture of highly bio available organic trace minerals that is intended to enhance the general performance of layers in means of bringing out uniformity in egg production, to prevent leg weakness, and to reduce the incidence of broken eggs or poor egg shell quality in layer farms.

COMPOSITION:

Optimized organic trace minerals : Zinc, Manganese, Copper, Iodine, Iron, Selenium, Chromium and Vitamin D3 from Plant origin to fulfill the nutrient requirement in layer birds.

1. Contribute to a robust flock with uniformity in terms of body size, contour and production.
2. Maintains the strength, quality, and homogeneity of the eggshell, egg size, bone, and foot pad.
3. Enhances both internal (Albumen and yolk) and external (Shell) quality of eggs.
4. Protect the flock from chondrodystrophy and perosis.
5. Enrich the immunity and develop disease resistance.
6. Enhanced skin and intestinal integrity.

Dosage:

500-750 gm / tonne of feed.

Presentation:

25 kg bag.

Storage:

Store in a cool, dry place.

Keep the bag closed when not in use.



(An ISO 9001 : 2015 Certified Company)



GMP & HACCP Certified

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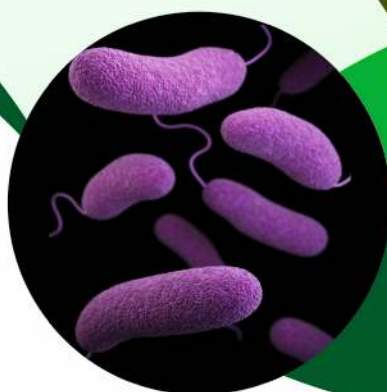
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ZenAct-PRO



ZenAct-Pro : Advanced Gut Health and Performance Booster for Poultry

Our innovative poultry supplement, ZenAct-Pro is specifically formulated to enhance gut health, improve intestinal integrity and optimize overall poultry performance. By targeting key factors responsible for gut health, ZenAct-Pro effectively reduces loose droppings and minimizes the percentage of dirty eggs, ensuring improved hygiene, better productivity and increased profitability at the farm level.

COMPOSITION

This advanced formulation combines naturally derived gut-acting probiotic strains, an epithelial cell enhancer that fortifies the intestinal lining, SCFAs, MCFAs, Mannans, Catalytic proteins, NSPs digesting enzymes and Astringents.

1. By promoting the growth of beneficial bacteria and inhibiting harmful pathogens, the gut-acting probiotic strains in ZenAct-Pro ensure a balanced microbiome.
2. The epithelial cell enhancer and SCFAs work together to reinforce the integrity of the intestinal lining. A strong intestinal wall prevents harmful pathogens from passing through, leading to healthier digestion, reduced inflammation and fewer cases of loose droppings.



Before using ZenAct-Pro



After using ZenAct-Pro for 10 days

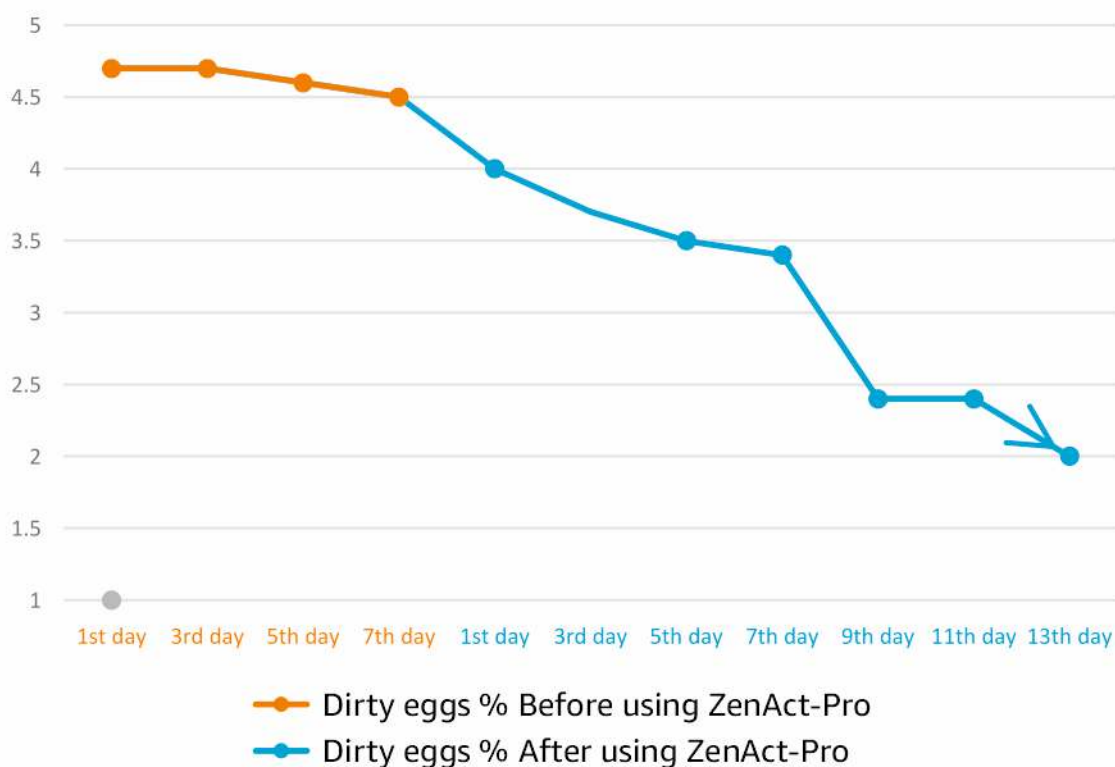
By controlling the wet litter problem, we also effectively reduce the fly population by inhibiting the growth of larvae. Dry litter creates an unfavorable environment for larvae, limiting their development and curbing fly infestation.

3. MCFAs are easily digestible fats that provide quick energy to the bird's system. They possess antimicrobial properties that target harmful bacteria in the gut, supporting the balance of gut flora and improving digestion.
4. Catalytic proteins and NSP digesting enzymes in ZenAct-Pro enhance metabolic and digestive processes by breaking down complex feed ingredients into more readily absorbable forms. This ensures better utilization of nutrients, reduces the amount of undigested feed in the gut and minimizes the risk of loose droppings due to poor digestion, while also lowering ammonia levels in poultry housing.
5. Astringents in ZenAct-Pro has moisture binding properties that ensure firmer droppings, leading to better litter conditions. With cleaner litter, hens are less likely to come into contact with contaminants, directly reducing the percentage of dirty eggs.



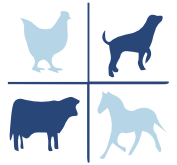
Trial report for Dirty Eggs %

Trial conducted in 64 weeks old layer flock



Dirty eggs % was recorded for 1 week before starting trial with ZenAct-Pro.

Decrease in Dirty eggs % was noticed after 2 weeks of supplementing ZenAct-Pro in feed.



BENEFITS:

- Promotes better digestion and nutrient absorption, leading to firmer droppings and overall healthier birds.
- Decreased incidence of dirty eggs results in higher market value and less wastage, improving overall farm profitability.
- Healthier birds with better digestion lead to more consistent egg production and improved growth rates, enhancing overall farm productivity.

DOSAGE:

During severe Loose droppings / Dirty eggs conditions – 2 kg / Tonne of feed for 10 days.

For regular use – 1 kg / tonne of feed

or As advised by the Veterinarian

** 2 kg/tonne of feed - Total replacement of Enzymes and Probiotics in feed.

1kg/tonne of feed - Replacement of half the dose of Enzymes and Probiotics in feed.

PRESENTATION:

25 KG Bag

SHELF LIFE:

24 months from the date of manufactured.



(An ISO 9001 : 2015 Certified Company)



GMP & HACCP Certified

Manufactured and Marketed By :

ZENMAK NUTRIGENCIES AND HEALTH PVT. LTD.

Regd. Off: No. F2, Sand Stone Apartments,
14/5, Main Road, Velacherry, Chennai - 600 042.

E-mail : info@zenmakglobal.com

Contact : 93459 13689, 93449 31464

Web : www.zenmakglobal.com





Perfom-X

Unlock the potential



Layer and Breeder birds play a vital role in the poultry industry by supplying nutritious eggs to meet the dietary needs of consumers worldwide and for producing fertile eggs that hatch into healthy chicks.

Introducing the groundbreaking field of nutrigenomics in poultry production: Perfom-X—a cutting-edge approach that combines the power of nutrition and genetics to optimize the health and performance of poultry.

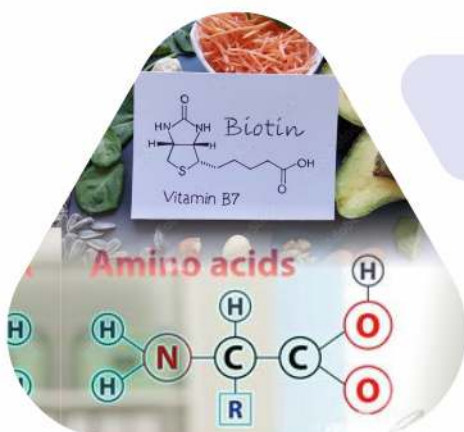
Perfom-X contains a precise blend of Marine algae, Yeast nucleotide concentrate and Micro and Macro nutrient blend tailored to meet the specific needs of laying hens. This scientifically optimized formula ensures that hens receive the ideal nutritional support to thrive and excel in egg production.

Nucleotides are the building blocks of DNA and RNA, essential for cell division, growth and reproduction. During egg formation, there is a high demand for the synthesis of DNA and RNA in developing ova (eggs) and reproductive tissues such as the ovary and oviduct.



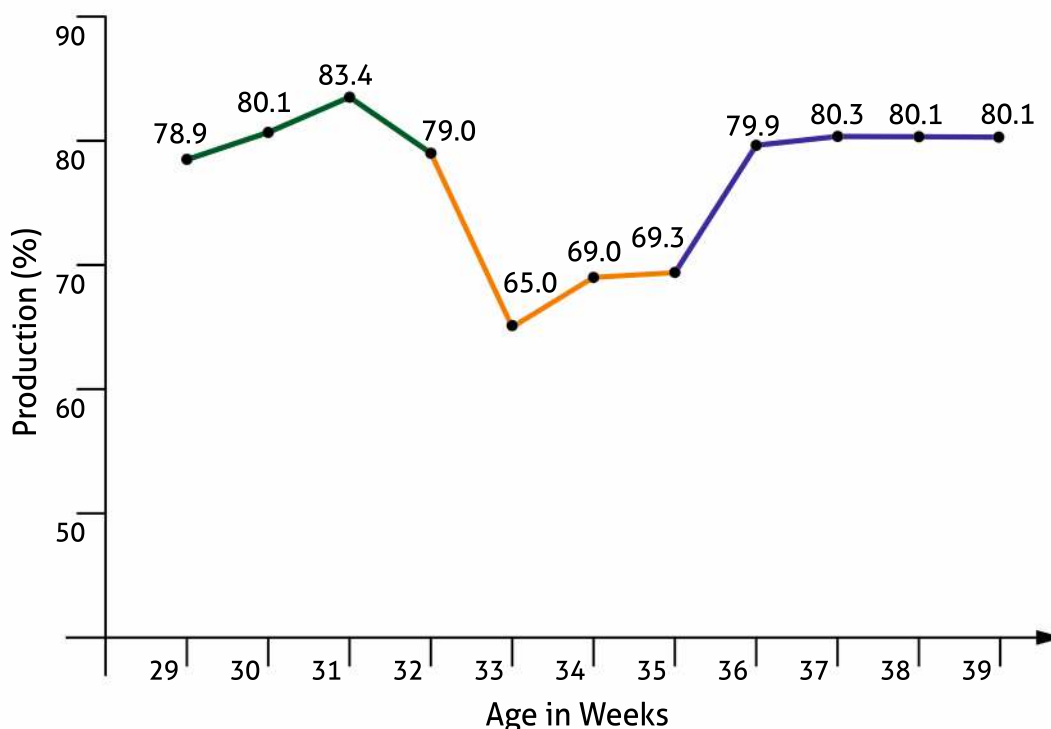
Nucleotides participate in ATP production that support the high metabolic demands associated with egg production, including follicular development, ovulation and eggshell formation.

Improvement in Chick quality and yolk weight due to Linoleic acid content. The bioactive compounds found in seaweed algae are potent Antioxidants: Seaweed algae contain antioxidants such as polyphenols, flavonoids, and carotenoids. These compounds help in reducing oxidative stress in poultry birds, which can positively impact reproductive performance and egg production.

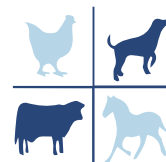


Perform-X is packed with essential nutrients like vitamins, minerals and amino acids, which are crucial for maintaining reproductive health.

TRIAL CONDUCTED IN ILT AFFECTED BREEDER FLOCK NEAR PALLADAM IN TAMIL NADU.



- Production % before ILT infection in the flock.
 - Production % drop due to ILT infection
 - Production % after supplementing PERFOM-X @ the dosage of 2 KG / tonne of feed
- The flock was treated with antibiotics for ILT infection from 33.2 to 34.1 weeks of age.



BENEFITS:

- Increase the growth of follicles and the generation of eggs.
- Helps for the rapid cell division required during egg formation.
- Achieve Higher Hatchability Rates for Optimal Yield and Profitability.
- Revitalize the laying potential with potent antioxidants.

DOSAGE:

Layer, breeder: 500 g / tonne of feed.

Recovery Phase: 1 - 2 KG /tonne of feed.

(or)

As advised by the Veterinarian / Nutritionist.

PRESENTATION:

25 KG

SHELF LIFE:

24 months from the date of manufacture.

*"Fortify the Potential :
Elevate Egg Production and Boost
Profitability with Our Innovative Solutions!"*



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